Nazorat testi · *Aralash test · 18-variant* — *Milliy sertifikat 2026. Kimyo*

Vaqt: *180 daqiqa*

Davriy jadvaldan foydalanish mumkin.

1. $N_2O_4 \rightleftarrows 2NO_2$ reaksiyasi 1 l hajmli idishda o‘tkazildi. Ma’lumotlar jadvalda berilgan:

Modda	N_2O_4	NO_2
Boshlang‘ich, mol	2	0
Muvozanatda, mol	—	1

Muvozanat **konstantasini (K) hisoblang.**
A) *1,333* B) *0,333* C) *0,667* D) *1,5*

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2. 40 g NaOH 500 g suvda eritildi. Eritmaning molyal **konsentratsiyasini (mol/kg) aniqlang.**
A) *1,85* B) *0* C) *2* D) *4*

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3. NH_4NO_3 tarkibida N ning oksidlanish **darajasini aniqlang.**
A) *+6* B) *+5* C) *−3 va +5* D) *+3*

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4. 1 mol HNO_3 va 3 mol $Ba(OH)_2$ suvda eritilib, eritma hajmi 2 l ga yetkazildi. Eritmadagi barcha ionlarning umumiy molyar **konsentratsiyasini (mol/l) aniqlang** (kuchli elektrolitlar to‘liq dissotsilanadi).
A) *5,5* B) *2* C) *11* D) *2,75*

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5. $N_2 + 3H_2 \rightleftarrows 2NH_3$ reaksiyada muvozanat qaror topganda $[N_2] = 1$, $[H_2] = 1$, $[NH_3] = 4$ mol/l bo‘ldi. N_2 ning boshlang‘ich **konsentratsiyasini (mol/l) aniqlang.**
A) *1* B) *3* C) *5* D) *6*

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6. Jadvalda $NaNO_3$ ning ikki eritmasi haqida ma’lumot berilgan:

Eritma	Massasi, g	Massa ulushi, %
1	250	10
2	100	25

Eritmalar aralashtirilib, ustiga 50 g suv qo‘shildi. Hosil bo‘lgan eritmaning massa **ulushini (%) aniqlang.**
A) *17,5* B) *35* C) *20,71* D) *12,5*

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7. H_2 va CH_4 dan iborat 10,08 l (n.sh.) aralashmaning massasi 4,4 g. Aralashmadagi H_2 ning **massasini (g) aniqlang.**
A) *0,4* B) *0,5* C) *3,2* D) *4*

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8. 36,5 g HCl 200 g suvda eritildi. Eritmaning molyal **konsentratsiyasini (mol/kg) aniqlang.**
A) *4,23* B) *0,01* C) *10* D) *5*

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9. Faqat qaytaruvchi bo‘la oladigan **ionlarni aniqlang**: 1) Zn^{2+}; 2) Na^+; 3) K^+; 4) S^{2-}
A) *2* B) *3* C) *4* D) *3, 4*

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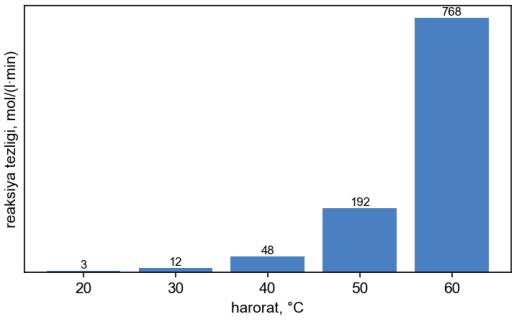
10. $H_2N–CH_2–COOH$ molekulasidagi σ va π bog‘lar **sonini aniqlang.**
A) *8 va 1* B) *9 va 2* C) *10 va 1* D) *9 va 1*

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11. O_2 gazining -23 °C va 2 atm bosimdagi **zichligini (g/l) aniqlang.** (R = 0,082 l·atm/(mol·K))
A) *6,24* B) *1,56* C) *3,12* D) *1,43*

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12. Reaksiya tezligining haroratga bog‘liqligi diagrammada ko‘rsatilgan.



Reaksiyaning temperatura **koeffitsiyentini (γ) aniqlang.**
A) *4* B) *5* C) *3* D) *8*

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13. K_2CO_3 ning suvli eritmasi inert elektrodlar bilan elektroliz qilinganda qaysi moddalar ajraladi?
A) *H_2 va O_2* B) *Cu va Cl_2*
C) *Zn, H_2 va O_2* D) *H_2, Cl_2 va NaOH*

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14. Qaysi juftliklar o‘zaro reaksiyaga kirishmaydi? 1) Fe_2O_3 va H_2O; 2) SiO_2 va H_2O; 3) Zn va NaOH; 4) NaCl va HNO_3
A) *3, 4* B) *1, 4* C) *1, 2, 4* D) *1, 2, 3*

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15. $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$ jarayoni uchun tarkibida 50% aralashma bo‘lgan 128 g namuna (asosiy modda Cu) olingan. $Cu(NO_3)_2$ **massasini (g) aniqlang.**
A) *376* B) *94* C) *282* D) *188*

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16. $CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$ reaksiyasida 2,24 l (n.sh.) CO_2 ishtirok etdi. $CaCO_3$ **massasini (g) toping.**
A) *20* B) *10* C) *5* D) *15*

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17. $NaBr + AgNO_3 \rightarrow AgBr + NaNO_3$ reaksiyasida tarkibida 60% aralashma bo‘lgan 103 g namuna (asosiy modda NaBr) ishtirok etdi. AgBr **massasini (g) toping.**
A) *150,4* B) *37,6* C) *112,8* D) *75,2*

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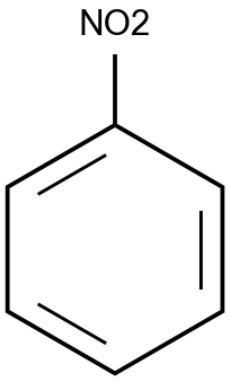
18. Ishqoriy metallar haqida qaysi fikrlar to‘g‘ri? 1) Kaliy suvdan og‘ir; 2) KOH – kuchli ishqor; 3) Ishqoriy metallar faqat oksidlovchi bo‘ladi; 4) Natriy suv bilan reaksiyada H_2 va NaOH hosil qiladi
A) *2, 3* B) *2, 4* C) *1* D) *1, 2, 4*

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19. Alyuminiy va magniy dan iborat 10,5 g aralashma mo‘l NaOH eritmasida eritilganda 10,08 l (n.sh.) gaz ajraldi. Aralashmadagi magniy ning massa **ulushini (%) aniqlang.**
A) *77,1* B) *22,9* C) *11,4* D) *69,4*

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20. Rasmda aromatik uglevodorod hosilasining tuzilish formulasi berilgan.



Molekuladagi σ-bog‘lar **sonini aniqlang** (π-bog‘lar halqada va o‘rinbosarda alohida hisoblanadi).
A) *14* B) *15* C) *13* D) *16*

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21. $CH_3–CH=C(CH_3)–CH_3$ birikmaning sistematik nomi qaysi qatorda to‘g‘ri keltirilgan?
A) *3-metilbuten-2* B) *2-metilbuten-3*
C) *2-metilbuten-2* D) *2-metilbuten-1*

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22. 15,6 g benzoldan nitrobenzol olishda unum 80% bo‘lsa, mahsulot **massasini (g) aniqlang** (boshlang‘ich modda massasi 15,6 g).
A) *19,68* B) *24,6* C) *39,36* D) *12,48*

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23. CH_3OH va C_3H_7OH dan iborat 15,2 g aralashma natriy bilan to‘liq reaksiyaga kirishganda 3,36 l (n.sh.) vodorod ajraldi. Aralashmadagi CH_3OH **massasini (g) toping.**
A) *12* B) *6* C) *15,2* D) *3,2*

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24. 22 g CH_3CHO dan quyidagi ikki bosqichda $CH_3COOC_2H_5$ olinadi: 1) $CH_3CHO + 2Cu(OH)_2 \rightarrow CH_3COOH + Cu_2O + 2H_2O$ (unum 60%); 2) $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ (unum 80%). Oxirgi mahsulot $CH_3COOC_2H_5$ ning **massasini (g) toping.**
A) *35,2* B) *26,4* C) *44* D) *21,12*

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25. Karbon kislotalar haqida qaysi fikrlar to‘g‘ri? 1) Sirka kislota karbonatlar bilan CO_2 ajratadi; 2) Sirka angidrid ikki molekula sirka kislotadan suv chiqib hosil bo‘ladi; 3) Chumoli kislota "kumush ko‘zgu" reaksiyasini beradi; 4) Karboksil guruh karbonil va gidroksil guruhlardan iborat
A) *1, 3, 4* B) *1, 2, 3, 4*
C) *1, 2, 4* D) *2, 3, 4*

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26. 22 g $C_2H_5COOCH_3$ mo‘l NaOH bilan to‘liq gidrolizlandi. Hosil bo‘lgan spirt **massasini (g) aniqlang.**
A) *22* B) *16* C) *8* D) *24*

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27. Uglevodlar haqida qaysi fikrlar to‘g‘ri? 1) Selluloza β-glyukoza qoldiqlaridan iborat; 2) Glyukoza "kumush ko‘zgu" reaksiyasini beradi; 3) Kraxmal yod bilan ko‘k rang beradi; 4) Saxaroza Feling suyuqligini qaytaradi
A) *1, 3* B) *1, 2, 3* C) *1, 2, 4* D) *2, 3*

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28. Aminlar, aminokislotalar va oqsillar haqida qaysi fikrlar to‘g‘ri? 1) Oqsillar biuret reaksiyasini beradi; 2) Aminokislotalar amfoter birikmalar; 3) Dipeptidda 2 ta peptid bog‘ bor; 4) Glitsin optik faol
A) *1, 2* B) *2, 3, 4* C) *2, 3* D) *4*

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29. Reaksiya tenglamasi: $C_2H_4 + H_2 \rightarrow C_2H_6$. Reaksiyaga 22,4 l (n.sh.) C_2H_4 kirishdi. C_2H_6 **hajmini (l, n.sh.) hisoblang.**

A) *44,8* B) *22,4* C) *11,2* D) *33,6*

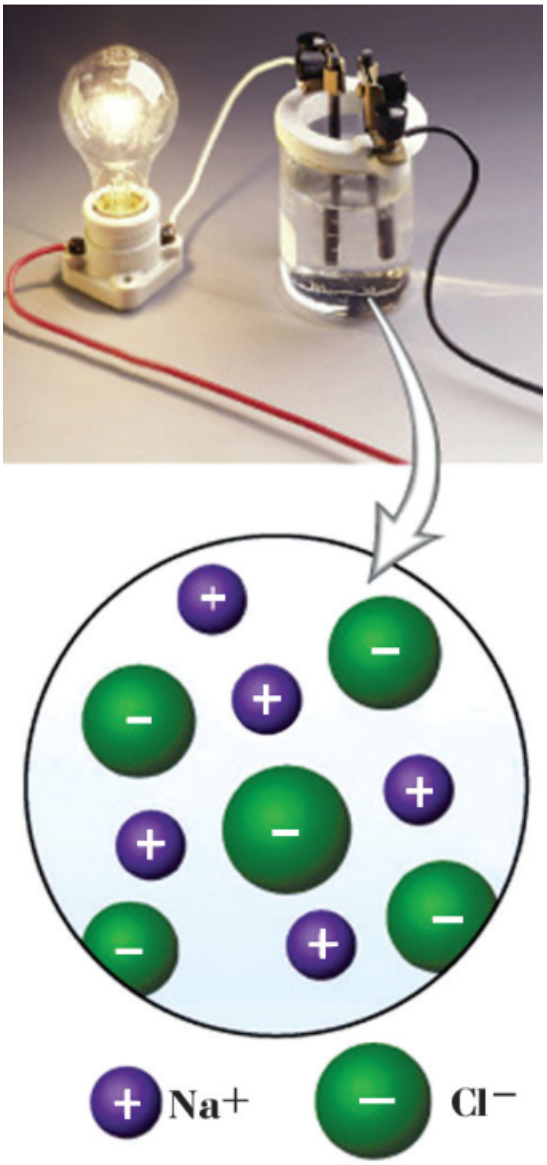
30. Metiloranj indikatori kislotali muhitda qanday rang beradi? A) *qizil*

B) *malina rangga kiradi (pushti)*

C) *rangsiz*

D) *binafsha*

31. Rasmdagi asbobning elektrodlari quyidagi eritmalardan qaysi biriga tushirilganda lampochka yonadi?



A) *distillangan suv* B) *mis sulfat eritmasi*
C) *shakar eritmasi* D) *etanol*

32. Kimyo laboratoriyasida ishlash qoidalari haqida qaysi fikrlar to‘g‘ri? 1) Suvni konsentrlangan kislotaga tez quyish mumkin; 2) Ishqor teriga tushsa, qo‘l suv bilan yuviladi; 3) Kimyoviy moddalar hidini bevosita nafas olib aniqlash kerak; 4) Spirt lampa boshqa lampadan yoqilmaydi
A) *3* B) *1, 4*
C) *2, 3, 4* D) *2, 4* —

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Topshiriqlar (33–35) va javob variant (A–F)larini o‘zaro to‘g‘ri moslashtiring. NaI va KBr dan iborat 26,9 g aralashma suvda eritilib, mo‘l $AgNO_3$ eritmasi qo‘shildi; hosil bo‘lgan cho‘kmalarning umumiy massasi 42,3 g.	A *30* B *18,6* C *14,88* D *15* E *340* F *22,5*
33. Aralashmadagi NaI massasini (g) aniqlang.	
34. Sarflangan 10% li $AgNO_3$ eritmasining massasini (g) aniqlang ($M(AgNO_3) = 170$ g/mol).	
35. Eritmada hosil bo‘lgan nitratlarning umumiy massasini (g) aniqlang.	

36. $KMnO_4 + Na_2SO_3 + H_2O \rightarrow MnO_2 + Na_2SO_4 + KOH$ reaksiya tenglamasini tenglang. Reagentlar (chap qism) koeffitsiyentlari **yig‘indisini toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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37. 1 l hajmli yopiq idishda 1 mol N_2O_4 qizdirilganda $N_2O_4 \rightleftarrows 2NO_2$ muvozanati qaror topdi. Muvozanat aralashmasining vodorodga nisbatan zichligi 38,33 ga teng. Shu haroratdagi muvozanat **konstantasini (K, mol/l) hisoblang.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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38. $CaCO_3$ va $MgCO_3$ dan iborat 26,8 g aralashma doimiy massagacha qizdirilganda qoldiq massasi 13,6 g bo‘ldi. Aralashmadagi $CaCO_3$ ning **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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39. C_3H_7OH va CH_3OH dan iborat 9,2 g aralashma natriy bilan to‘liq reaksiyaga kirishganda 2,24 l (n.sh.) vodorod ajraldi. Aralashmadagi C_3H_7OH **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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40. 22,2 g propion kislota va 9,6 g metanol dan 75% unum bilan murakkab efir olindi. Efir **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

YOZMA ISH

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41. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

Sulfat kislotaning massa ulushi 40% bo‘lgan eritmasining zichligi 1,3 g/ml.
1)Eritmaning molyar konsentratsiyasini (mol/l) hisoblang.
2)Eritmaning normal konsentratsiyasini (ekv/l) toping. Sulfat kislotaning ekvivalent massasini (g/ekv) ham aniqlang.
3)Eritmaning molyal konsentratsiyasini (mol/kg) aniqlang.
4)Shu eritmadan 250 ml 0,2 mol/l li eritma tayyorlash uchun necha ml olish kerak? 100 ml eritmadagi suvning massasini (g) ham hisoblang.
5)100 ml shu eritmani to‘liq neytrallash uchun necha gramm NaOH kerak? 250 ml shu eritmadagi H_2SO_4 massasini (g) ham hisoblang.

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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42. Reaksiya tenglamalarini tugallang va tenglang. Hosil bo‘lgan organik moddalarning tuzilish formulalarini yozing va ularni nomlang.
1)n $CH_2{=}CH{-}CH{=}CH_2 \rightarrow$
2)$CH_3COONa + NaOH \xrightarrow{t}$
3)$C_2H_5OH \xrightarrow{H_2SO_4(\text{kons.}),\ 170\ ^\circ C}$
4)$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+}$
5)$CH_3CHO + Ag_2O \xrightarrow{NH_3,\ t}$
6)$H_2N{-}CH_2{-}COOH + NaOH \rightarrow$
7)$C_6H_5{-}CH_3 + Cl_2 \xrightarrow{\text{yorug‘lik}}$

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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43. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

120 g pirit (FeS_2) yondirildi: $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$ (unum 80%). Hosil bo‘lgan SO_2 katalizator ishtirokida SO_3 gacha oksidlandi (unum 80%), SO_3 suvda eritildi.
1)Hosil bo‘lgan SO_2 hajmini (l, n.sh.) va qattiq qoldiq Fe_2O_3 massasini (g) hisoblang. Shuningdek SO_2 massasini (g) ham toping.
2)SO_3 ning massasini (g) toping ($2SO_2 + O_2 \rightleftarrows 2SO_3$).
3)Hosil bo‘lgan H_2SO_4 massasini (g) aniqlang.
4)Shu kislotani to‘liq neytrallash uchun zarur NaOH massasini (g) va mo‘l $BaCl_2$ bilan hosil bo‘ladigan cho‘kma massasini (g) toping.
5)Shu H_2SO_4 dan 20% li eritma tayyorlash uchun qancha suv (g) kerak?

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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Nazorat testi · *18-variant* **Javoblar kaliti**

Faqat o‘qituvchi uchun.

1–32-savollar

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
C	*C*	*C*	*A*	*B*	*D*	*A*	*D*	*C*	*D*	*C*	*A*	*A*	*C*	*D*	*B*
17	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**
D	*B*	*B*	*A*	*C*	*A*	*D*	*D*	*B*	*C*	*B*	*A*	*B*	*A*	*B*	*D*

33–35-savollar

No	Javob	Qiymat
33	*D*	15
34	*E*	340
35	*B*	18,6

36–40-savollar

No	To‘g‘ri javob
36	6
37	0,2
38	10
39	6
40	19,8

41–43-topshiriqlar: javoblar

41-topshiriq.

1) 5,306 mol/l
2) 10,612 ekv/l; 49 g/ekv
3) 6,803 mol/kg
4) 9,42 ml; 78 g
5) 42,45 g; 130 g

42-topshiriq.

1) n $CH_2{=}CH{-}CH{=}CH_2 \rightarrow (-CH_2-CH{=}CH-CH_2-)_n$ — polibutadiyen (3 ball)
2) $CH_3COONa + NaOH \xrightarrow{t} CH_4 + Na_2CO_3$ — metan (4 ball)
3) $C_2H_5OH \xrightarrow{H_2SO_4(kons.),\ 170\ ^\circ C} CH_2{=}CH_2 + H_2O$ — etilen (3 ball)
4) $C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$ — glyukoza va fruktoza (4 ball)
5) $CH_3CHO + Ag_2O \xrightarrow{NH_3,\ t} CH_3COOH + 2Ag\downarrow$ — etan (sirka) kislota (4 ball)
6) $H_2N{-}CH_2{-}COOH + NaOH \rightarrow H_2N{-}CH_2{-}COONa + H_2O$ — glitsinning natriyli tuzi (3 ball)
7) $C_6H_5{-}CH_3 + Cl_2 \xrightarrow{\text{yorug‘lik}} C_6H_5CH_2Cl + HCl$ — benzilxlorid (4 ball)

43-topshiriq.

1) 35,84 l; 64 g; 102,4 g
2) 102,4 g
3) 125,44 g
4) 102,4 g; 298,24 g
5) 501,76 g

Mavzular taqsimoti

No	Mavzu	Tekshiriladigan ko‘nikma	Ball
1	Tirkashev VI bob (muvozanat, jadval)	Jadval tahlili	1
2	11-sinf 4-bob (molyal konsentratsiya)	Hisoblash va masala yechish	1
3	8-sinf; Tirkashev *V* bob (oksidlanish darajasi)	Bilim va tushunish	1
4	Tirkashev IV bob (kuchli elektrolitlar)	Hisoblash va masala yechish	1
5	Tirkashev VI bob	Hisoblash va masala yechish	1
6	Tirkashev III bob (eritmalar, jadval)	Jadval tahlili	1

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7	11-sinf (gaz aralashmalar)	Hisoblash va masala yechish	1
8	11-sinf 4-bob (molyal konsentratsiya)	Hisoblash va masala yechish	1
9	Tirkashev *V* bob; 11-sinf 7-bob	Bilim va tushunish	1
10	10-sinf §17 (gibridlanish); Abdusamatov	Bilim va tushunish	1
11	11-sinf (Mendeleyev–Klapeyron)	Hisoblash va masala yechish	1
12	Tirkashev VI bob (Vant-Goff qoidasi, grafik)	Grafik tahlili	1
13	11-sinf 8-bob; Tirkashev *V* bob (elektroliz)	Bilim va tushunish	1
14	9-sinf; Tashev (anorganik birikmalar xossalari)	Bilim va tushunish	1
15	9-sinf; Tirkashev ($N/NH_3/HNO_3$, hisob)	Hisoblash va masala yechish	1
16	9-sinf; Tirkashev (P/C/Si, hisob)	Hisoblash va masala yechish	1
17	9-sinf; Tirkashev (galogenlar, hisob)	Hisoblash va masala yechish	1
18	9-sinf (ishqoriy metallar); Tashev	Bilim va tushunish	1
19	9-sinf (alyuminiy)	Hisoblash va masala yechish	1
20	10-sinf (arenlar)	Rasm bo‘yicha tahlil	1
21	10-sinf (nomenklatura §6, §11, §20, §24–26)	Bilim va tushunish	1
22	10-sinf §16, §23, §33	Hisoblash va masala yechish	1
23	10-sinf §21	Hisoblash va masala yechish	1
24	10-sinf (ikki bosqichli hisob)	Hisoblash va masala yechish	1
25	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
26	10-sinf §28	Hisoblash va masala yechish	1
27	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
28	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
29	10-sinf (pol, hisob)	Hisoblash va masala yechish	1
30	8-sinf (indikatorlar)	Bilim va tushunish	1
31	9-sinf (elektrolitlar, 1-rasm)	Rasm bo‘yicha tahlil	1
32	7-sinf (xavfsizlik qoidalari); 10-sinf lab.	Bilim va tushunish	1
33–35	Aralash masala (moslashtirish)	Ko‘p bosqichli hisob	3
36–40	Qisqa javobli masalalar	Hisoblash	5
41–43	Yozma ish	Usul va hisob (M, A)	75

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